

{EPITECH}

DR CVRIE

< CONTAGIONS VULNERABILITY ROBOTIC
INFIRMARY ENGINEER />



DR CVRIE



binary name: *.py, *.ipynb

language: Python

compilation: none

Authorized tools: Scikit-learn, Pandas, Matplotlib, Numpy, Jupyterlab

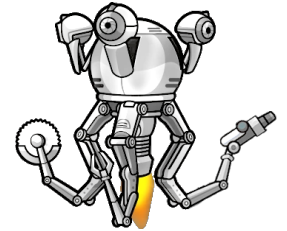


- ✓ The totality of your source files, except all useless files (binary, temp files, objfiles,...), must be included in your delivery.
- ✓ All the bonus files (including a potential specific Makefile) should be in a directory named bonus.
- ✓ Error messages have to be written on the error output, and the program should then exit with the 84 error code (0 if there is no error).

CVRIE

Meet CVRIE. They're an artificial intelligence specialized in medicine, capable of performing various tasks. They can analyze images, care for patients, and make diagnoses, all with great precision and efficiency. At least they should, but they're not completely autonomous yet. We need your help.

I'm CVRIE V0, I DO exist, and I'm perfect. Every patient I've had has stopped being ill after my care!



CVRIE is not perfect. I can't stress that enough. Their integration in our medical teams has led to catastrophic results in our early trials.

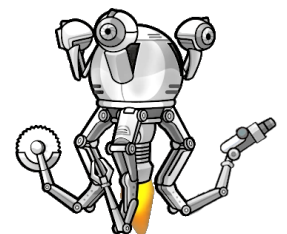
Your goal for this project is to develop CVRIE V1, a model smart enough to handle two of the most important tasks in this hospital:

- Analyzing **incoming patient testimonies** and grouping them by pathology or degree of urgency
- Analyzing **medical images** and scans in order to formulate a diagnosis

We need this to work. The health of a lot of people depends on you. You **must** achieve the best results, by any means necessary. But, you see, if I told that to CVRIE themselves, who knows what "means" they would use... So, in order to protect this hospital's reputation, and keep ourselves out of jail, you'll have to do everything by the books and provide an **in-depth analysis of all of your decisions and insights**. Are we clear?

Do you have the guts to do it even if that means working alongside them?

My scanners detect a 50 percent chance of guts inside of this student. I would recommend **crystals** and **radioactivity**. Vital prognostic engaged.



Please help us.

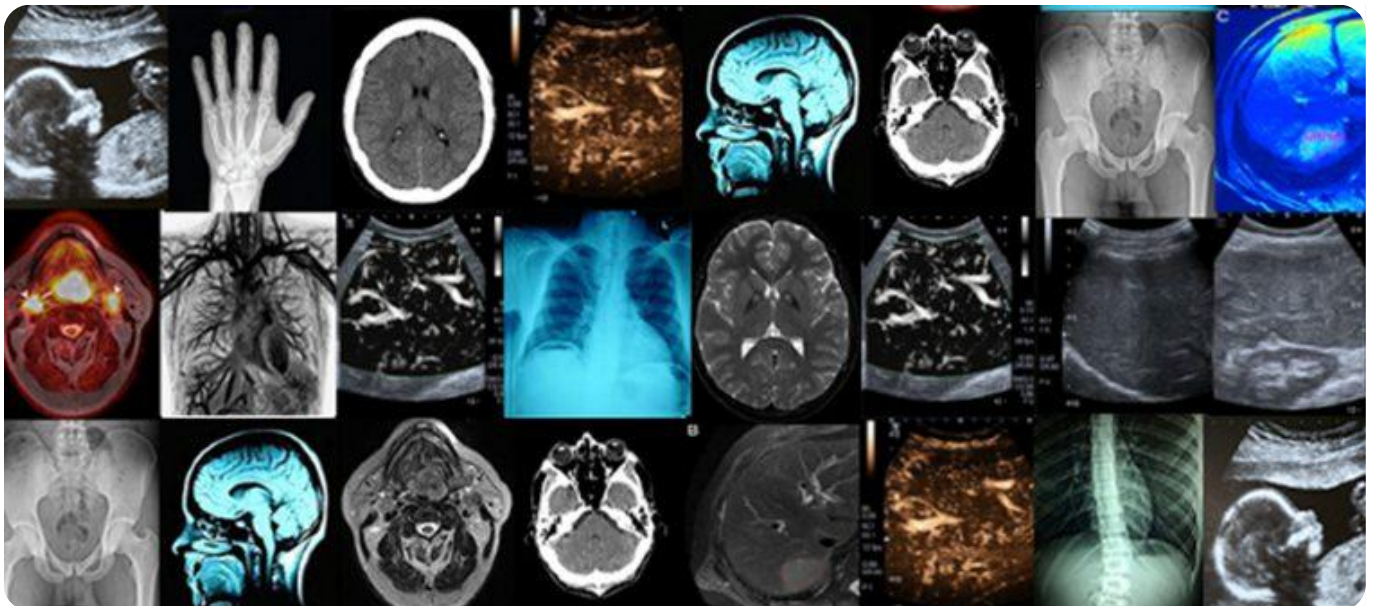
Supervised Learning

Dataset

The CVRIE's first assignment will require **supervised training**. You will choose a medical imaging dataset and train a model that generates the best predictions for it.

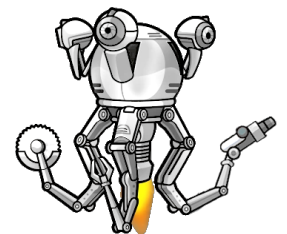
This dataset must obey some constraints in order to be considered valid for this project:

- The dataset only contains images and their associated labels. You may trim additional information. Your model mustn't take them into account.
- The overall dataset is representative of a **classification** issue.
- Your whole group feels comfortable with the dataset's theme.
- You are able to justify your choice. Choosing the same dataset as another group without justification is forbidden.



Your notebook for this task must contain an analysis of your dataset. Interactive visualisation of the dataset's characteristics could be a good starting point in identifying flaws within the data that you could then compensate for in the next step.

In order for me to do my job well and for there to be no more illness in the whole world, please select a dataset with no illness. Thank you.



Cleaning up the data and Pre-Processing

Images Classification Datasets can be biased or otherwise imperfect in different ways:

- Your data could need to be **normalised**. Images of different sizes, color palettes or resolution might need some pre-processing.
- Your data could be **unbalanced**. Are positive and negative cases both present and equally represented in your data?
- Are there **outliers** that skew the data in specific directions or extremes?



If your data is composed of 90% positive cases, a simple "return("Positive")" will be accurate 90% of the time.

Different problems with your dataset require the use of different tools. You are allowed to alter a dataset if you retrace your steps and reflection in your notebook.

Training and Evaluation

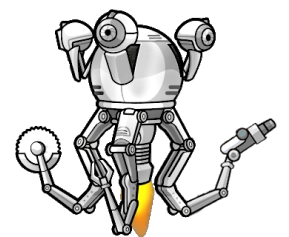
Machine learning works by fine-tuning predictions through multiple iterations of training, striving for better results each time. Each time, your model should try to adapt itself to reduce the value of its **loss function**. This concept should be part of your notebook. Selecting a good loss function is essential for your model's training to be efficient.



How do you split training data from evaluation data?

Your model will likely not be perfect from the get go. You should document your process and trial and error. Try using different algorithms, explore possibilities, and write down everything you did and your results. For this task, your model's result will only be a small part of what is evaluated. Your defense, where you will present your notebook, will be more important.

I love splitting stuff, can I do it please? Pretty please?



Unsupervised Learning

Dataset

CVRIE will work with patients and will sometimes be their first point of contact in the hospital. To streamline admission processes, it would be beneficial to be able to group patients by conditions and emergency. The hospital has provided you with a dataset of patient testimonials presenting various sets of symptoms. This dataset is not labeled. It will be up to you to train CVRIE to detect similarities between the testimonials. CVRIE is expected to "sort" them in categories or **clusters**.

That said, this information (patient conditions) exists even if it is not communicated to you. Your CVRIE's clusters should make sense in that they are as close as possible to how a real human would regroup those patients.

You are not allowed to modify the proposed dataset at all (by adding your own labels, for example).

Here is a small example of data from the dataset :

ID	Testimony
42	"My ears felt full and my nose was runny, but I didn't consider it a cold. A colleague told me it was typical of a common cold."
43	"I've been increasing my runs and now have a pinpoint ache on the inside of my shin. It's tender to touch and hurts even during easy jogs, worse on hard pavement."
44	"My sister told me I cough a lot at night, and it seems like I can't catch my breath in the middle of the night."

Cleaning up the data and Pre-Processing

You **must** include and document a pre-processing step for this task.

A number of the lessons learned in the first task will be applicable to this one



This specific problem might have a name that would lead you to more specific steps you could take?

Training and Evaluation

As with the first task, you will be evaluated both through your results and your notebook. Since the dataset doesn't have any label, you'll have to figure out another way to monitor your progress and compare your models. This will mostly be done through your data's features and choosing which to prioritize. You can try to label a small portion of the dataset by yourself, but you **mustn't** feed this information to your models, as that would defeat the purpose.

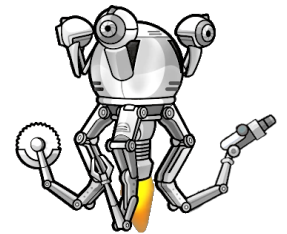
You will notice a column with a lot of "0x000000" in your dataset. When plotting your clusters, please use it to color the data. During your defense, the "real" color will be revealed, and your examiner should be able to see if your clusters are coherent at a glance.

Features Checklist

Here is a clearer list of things we've mentioned in this subject that will be looked for during the evaluation of this project. This list is not exhaustive, complete, or a substitute for your own exploration, reading, extrapolating and choices.

- ✓ A notebook for each part
- ✓ A python code that outputs clustering of the data in the unsupervised part of the project
- ✓ The dataset used for the supervised part of the project
- ✓ A python code that trains a model and classifies the data in the selected dataset in the supervised part of the project
- ✓ Clear section in each notebook for data pre-processing
- ✓ Clear section in each notebook for model comparisons
- ✓ Clear section in each notebook for conclusions and insight
- ✓ Clear explanation of the loss function in each notebook
- ✓ Beautiful charts and plots

I dislike the fact that this list is not exhaustive and complete or a substitute for thinking. Machine learning can't be about extrapolation, that's insane. You are insane. I recommend **tonsillectomy** and **staying out in the sun for hours without sunscreen**.



v 0.1

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